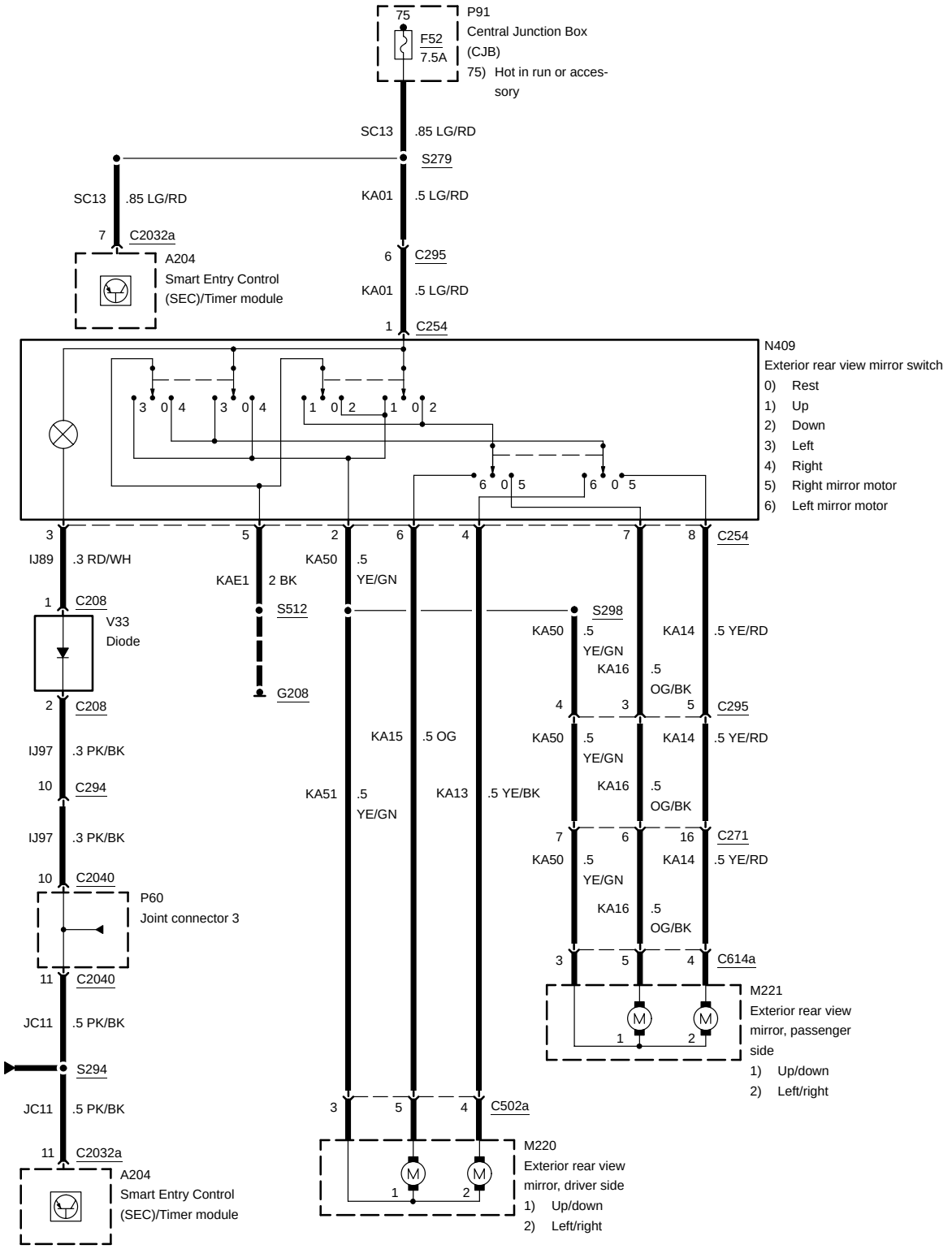without memory -1